An exact error floor for adaptive Monte Carlo approximation of a noncompact operator

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Abstract

For the canonical embedding $\ell_1 \hookrightarrow \ell_\infty$, we prove that the optimal worst-case root-mean-square error is exactly $1/2$ at every positive finite expected information budget. The proof permits adaptive continuous linear information and the joint measurability conventions of AIM 523. We also give compactness and separable-target consequences.

Catalogue target. AIM 523 at repository revision `c4036871b0bf`. The package preserves its full statement in `statement.md`; the review and target comparison are linked from the package README.

1 Problem 523: an exact error floor for adaptive Monte Carlo

Let $I : \ell_1 \to \ell_\infty$ be the identity inclusion. The input set is $B = \{x : \|x\|_1 < 1\}$. Measurements are continuous linear functionals, their choice and the stopping rule may depend arbitrarily on the preceding answers and the random seed, and the output and cardinality maps have the joint measurability required in the problem. The cost bound is $\sup_{x\in B} \mathbb{E}N(x,\omega) \le n$, and the error is the worst-input root mean square error. More explicitly, the seed ranges over an arbitrary probability space, $N(x,\omega)$ is finite and takes values in $\mathbb{N}\cup\{0\}$, and $(x,\omega) \mapsto N(x,\omega)$ and $(x,\omega) \mapsto A_\omega(x)$ are jointly measurable for the norm Borel structures and the seed sigma algebra. We write

$$e_n(I) = \inf_A \sup_{x\in B} \big(\mathbb{E}_\omega \|x - A_\omega(x)\|_\infty^2\big)^{1/2},$$

where the infimum is over those algorithms satisfying the stated expected cost bound.

Theorem 1.1. *In this model,*

$$\boxed{e_n(I) = \frac{1}{2} \qquad (n \ge 1).}$$

The issue left open in the sources is adaptivity, rather than the nonadaptive error floor; see [1, 2]. The proof below treats arbitrary adaptive decision rules. Its main ingredient is a tightness statement for the actual measurable output, which avoids imposing measurability on the rules selecting the queries.

*The solutions were found by AI. The submitted arguments and the repository audit are AI work; independent human review and formal verification are not claimed.

1.1 A tightness lemma for finite adaptive information

Lemma 1.2 (Uniform tightness under bounded shifts)**.** *Let E be a separable real Banach space, and let μ be a centered Radon Gaussian measure on E, with full support and finite second moment. Let $U \subset E$ be open. Suppose that $R : U \to \mathbb{N}_0$ is Borel measurable and is the output of an algorithm using at most k adaptive continuous affine scalar measurements.*

Let $V \subset E$ be countable and bounded, and let $w : E \to [0, \infty)$ be bounded and Borel measurable. Assume that, for some $\eta > 0$,

$$\operatorname{dist}(x + v, E \setminus U) \geq \eta \quad \text{whenever } w(x) > 0,\ v \in V. \tag{T1}$$

Extend R by zero outside U. Then

$$\lim_{M \to \infty} \sup_{v \in V} \int_E w(x) \mathbf{1}_{\{R(x+v) \geq M\}}\, d\mu(x) = 0. \tag{T2}$$

Neither the query-selection rules nor the decoder need be measurable as functions of the answer history.

Proof. We induct on k. With no measurements, the output is a fixed integer on U; (T1) then proves the assertion. An algorithm that stops early can be padded with zero measurements.

The first affine query has a fixed continuous linear part $a \in E'$. Its known additive constant may be subtracted from the answer. If $a = 0$, freeze this first answer and apply induction to the remaining queries. Otherwise, for $Z \sim \mu$, put

$$\sigma^2 = \mathbb{E}[a(Z)^2] > 0, \qquad c = \mathbb{E}[Z\, a(Z)] \in E, \qquad m(t) = \frac{c}{\sigma^2} t, \qquad E_0 = \ker a. \tag{T3}$$

The Bochner expectation exists by the second-moment assumption. Moreover $a(c) = \sigma^2$, so $a(m(t)) = t$. Gaussian regression gives

$$Z = m(Y) + Z_0, \qquad Y = a(Z), \tag{T4}$$

where Y has the $N(0, \sigma^2)$ density p, and Z_0 is an independent centered Gaussian random element of E_0. Indeed, every continuous linear functional of $Z - m(a(Z))$ has zero covariance with $a(Z)$; joint Gaussianity then gives independence. The law μ_0 of Z_0 has full support in E_0, because

$$P = \mathrm{Id} - m \circ a : E \to E_0$$

is a continuous surjection and μ has full support. Its second moment is finite. Thus the Gaussian hypotheses are preserved under conditioning.

For $v \in V$, write

$$b = a(v), \qquad v' = v - m(b) \in E_0. \tag{T5}$$

The scalar b and the residual shift v' range over bounded sets. Define the nonnegative integrable Borel function

$$f(y, z) = p(y) w(m(y) + z) \quad \text{on } \mathbb{R} \times E_0.$$

Changing the first-answer variable from t to $y = t + b$ gives

$$\begin{aligned} I_M(v) &:= \int_E w(x) \mathbf{1}_{\{R(x+v) \geq M\}}\, d\mu(x) \\ &= \int_{\mathbb{R}} \int_{E_0} f(y - b, z)\, \mathbf{1}_{\{R(m(y)+z+v') \geq M\}}\, d\mu_0(z)\, dy. \end{aligned} \tag{T6}$$

Fix $\varepsilon > 0$. Translation in the first real variable is continuous in $L^1(dy\, d\mu_0)$. Choose $h > 0$ such that

$$\|f(\cdot - t, \cdot) - f\|_{L^1} < \varepsilon \quad \text{and} \quad \|m(t)\|_E < \eta/2 \qquad (|t| < h). \tag{T7}$$

Since $a(V)$ is bounded, finitely many intervals of radius less than h, with centers $b_1, \dots, b_L$, cover it. Partition V into corresponding countable sets V_l, and put

$$W_l = \{v - m(a(v)) : v \in V_l\} \subset E_0. \tag{T8}$$

Each W_l is countable and bounded. For $v \in V_l$, replacing $f(y - a(v), z)$ in (T6) by $f(y - b_l, z)$ changes the integral by at most ε, uniformly in v and M.

Fix l and y, and define

$$\begin{aligned} U_y &= (U - m(y)) \cap E_0, \\ R_y(z) &= R(m(y) + z) \quad (z \in U_y), \\ w_{l,y}(z) &= w(m(y - b_l) + z). \end{aligned} \tag{T9}$$

The function R_y is Borel measurable. All inputs $m(y) + z$, $z \in E_0$, have the same first answer. Consequently R_y uses at most $k-1$ adaptive continuous affine measurements, with that answer frozen.

The shifted-support condition remains valid. If $w_{l,y}(z) > 0$, let $x = m(y - b_l) + z$. For $v \in V_l$, with v' as in (T5),

$$m(y) + z + v' = x + v + m(b_l - a(v)). \tag{T10}$$

By (T1) and (T7), this point has distance at least $\eta/2$ from $E \setminus U$. Distance within the affine subspace $m(y) + E_0$ is at least ambient distance. Hence

$$\operatorname{dist}_{E_0}(z + v', E_0 \setminus U_y) \geq \eta/2 \quad (w_{l,y}(z) > 0,\ v' \in W_l). \tag{T11}$$

Notice that the unshifted point $m(y) + z$ need not lie in U. This is why the hypothesis was formulated as (T1), rather than as a condition on the unshifted support alone.

Induction now implies, for every fixed y,

$$T_{M,l}(y) := \sup_{v' \in W_l} \int_{E_0} w_{l,y}(z) \mathbf{1}_{\{R_y(z+v') \geq M\}} \, d\mu_0(z) \longrightarrow 0. \tag{T12}$$

The supremum is countable. Each term is Borel measurable in y, directly from the Borel measurability of the original output R and of w. Thus $T_{M,l}$ is Borel, and $0 \leq T_{M,l} \leq \|w\|_\infty$. Dominated convergence gives

$$\int_{\mathbb{R}} p(y - b_l) T_{M,l}(y) \, dy \longrightarrow 0. \tag{T13}$$

There are only finitely many l. Combining (T6)–(T13) yields

$$\limsup_{M \to \infty} \sup_{v \in V} I_M(v) \leq \varepsilon.$$

Letting $\varepsilon \downarrow 0$ proves (T2). No history-to-query map has been integrated. □

1.2 Proof of the exact error floor

Proof of Theorem 1.1. For the upper bound, one measurement suffices:

$$s = \sum_{j \geq 1} x_j, \qquad A(x) = (s/2, s/2, \dots).$$

This is an admissible ℓ_∞-valued output, and

$$|x_j - s/2| = \frac{1}{2} \left| x_j - \sum_{i \neq j} x_i \right| \leq \frac{1}{2} \|x\|_1 . \tag{523.1}$$

For the lower bound, first impose a strict finite budget k. Let

$$Z_j = 2^{-j} g_j,$$

where the g_j are independent standard Gaussians. Then $Z \in \ell_1$ almost surely and $\mathbb{E}\,\|Z\|_1^2 < \infty$. Its law has full support in ℓ_1: a finite Gaussian head can approximate any prescribed finite vector with positive probability, and the independent tail can be made small with positive probability by first making its expected norm small.

Choose

$$0 < r < 1, \qquad 0 < \rho < \min\{r/2, 1-r\}, \qquad \tau = r/2 - \rho > 0. \tag{523.2}$$

Condition Z on $\|Z\|_1 \le \rho$, an event of positive probability. Equivalently, use the bounded weight

$$w(z) = \frac{\mathbf{1}_{\{\|z\|_1 \le \rho\}}}{\mathbb{P}(\|Z\|_1 \le \rho)}. \tag{523.3}$$

Take $U = B$ and $V = \{re_j : j \ge 1\}$. The margin in Lemma 1.2 is at least $1 - r - \rho > 0$.

For a fixed seed ω, define from the algorithm's output

$$R_\omega(x) = \min\{i \ge 1 : (A_\omega(x))_i > r/2\}, \tag{523.4}$$

with value zero if the set is empty. Coordinate evaluation is continuous on ℓ_∞, so R_ω is Borel. It uses the same measurements as A_ω. If $x = z + re_j$, $\|z\|_1 \le \rho$, and $\|x - A_\omega(x)\|_\infty < \tau$, then

$$(A_\omega(x))_j > r/2, \qquad (A_\omega(x))_i < r/2 \quad (i \ne j). \tag{523.5}$$

Thus $R_\omega(x) = j$. Uniform tightness implies

$$\mathbb{P}\big(R_\omega(Z + re_j) = j \mid \|Z\|_1 \le \rho\big) \longrightarrow 0, \tag{523.6}$$

because this probability is bounded by the uniformly vanishing tail at threshold j.

The probabilities in (523.6) are measurable functions of ω, by the joint measurability of the actual output. They are bounded by one. Dominated convergence therefore permits averaging (523.6) over any seed probability space. The norm error is jointly measurable even though ℓ_∞ is nonseparable: the map $x \mapsto Ix$ has separable range, so approximate it by countably valued Borel maps and take limits of their norm differences from the actual output. If e is the uniform RMS error, every shifted input law in (523.6) is supported in B, and hence

$$e^2 \ge \mathbb{E}_{Z,\omega}\big[\|Z + re_j - A_\omega(Z + re_j)\|_\infty^2 \mid \|Z\|_1 \le \rho\big] \ge \tau^2(1 - o(1)). \tag{523.7}$$

Let $j \to \infty$, then $\rho \downarrow 0$ and $r \uparrow 1$. This proves $e \ge 1/2$ for every strict finite budget.

For an expected budget, truncate the algorithm after k queries and return zero if it has not stopped. The truncated output is jointly Borel, since the original output and cardinality are jointly Borel. For every $x \in B$,

$$\begin{aligned} \mathbb{E}\left\|x - A_\omega^{(k)}(x)\right\|_\infty^2 &\le \mathbb{E}\,\|x - A_\omega(x)\|_\infty^2 + \|x\|_\infty^2\ \mathbb{P}(N(x,\omega) > k) \\ &\le e^2 + n/k. \end{aligned} \tag{523.8}$$

The strict-budget result gives $e^2 + n/k \ge 1/4$ for every k. Letting $k \to \infty$ completes the proof. □

Corollary 1.3 (If the output is required to belong to c_0)**.** *For the inclusion $I : \ell_1 \to c_0$, equipped with the same continuous-linear information, joint output/cardinality measurability, expected-budget convention, and worst-input RMS error,*

$$e_n(I : \ell_1 \to c_0) = 1 \qquad \textit{for every finite } n.$$

Proof. Fix $\varepsilon > 0$. For a deterministic realization with output $A(x) \in c_0$, the integer

$$R_A(x) = \max\{j \geq 1 : |A(x)_j| > \varepsilon\},$$

with value zero when the set is empty, is finite and Borel measurable. Indeed, $\{R_A \leq M\} = \bigcap_{j>M}\{|A(x)_j| \leq \varepsilon\}$. Use the same Gaussian law, conditioning, and shifts re_j as in the preceding proof. If $x = z + re_j$, $\|z\|_1 \leq \rho$, and

$$\|x - A(x)\|_\infty < r - \rho - \varepsilon,$$

then $|A(x)_j| > \varepsilon$, and therefore $R_A(x) \geq j$. Lemma 1.2 makes this event's conditioned probability tend to zero, for each seed, as $j \to \infty$. Dominated convergence over the seed gives an RMS lower bound $r - \rho - \varepsilon$ for every strict finite query budget. Let $\rho, \varepsilon \downarrow 0$ and $r \uparrow 1$. For expected budgets, truncation and zero output on unfinished paths give the same estimate $e^2 + n/k$ as in (523.8), so the lower bound remains one. The zero algorithm attains one. □

Remark 1.4. The value 1/2 for the ℓ_∞ codomain uses the constant sequence output of the one-query algorithm. A nonzero constant sequence does not belong to c_0; restricting the allowed output space therefore changes the exact randomized error.

1.3 A direct compactness criterion

Corollary 1.5. *Let $S : X \to Y$ be a bounded linear operator between real Banach spaces, with the same continuous-linear information, joint output/cardinality measurability, expected-budget convention, and worst-input RMS error. Then*

$$\lim_{n\to\infty} e_n(S) = 0 \quad \Longleftrightarrow \quad S \text{ is compact.} \tag{523.9}$$

More quantitatively, if there are $v_j \in X$, $\|v_j\|_X \leq 1$, such that $\|Sv_i - Sv_j\|_Y \geq \alpha > 0$ for $i \neq j$, then $e_n(S) \geq \alpha/2$ for every finite n.

Proof. Suppose first that S is noncompact and choose such a separated sequence. Its closed linear span $E \subset X$ is a separable Banach space. A full-support centered Gaussian with finite second moment exists on E. For example, if (u_l) lies in its unit ball and has dense linear span, the series

$$Z = \sum_{l\geq 1} 2^{-l} g_l u_l \tag{523.10}$$

converges absolutely almost surely and in $L^2(E)$; the finite-head and independent-tail argument used above proves full support.

Choose $0 < r < 1$, and then choose $\rho > 0$ so that

$$r + \rho < 1, \qquad \tau = r\alpha/2 - \|S\|\ \rho > 0.$$

Condition on $\|Z\|_X \leq \rho$ and apply Lemma 1.2 with the shifts rv_j. Classify the output by

$$R_A(x) = \min\{j : \|A(x) - rSv_j\|_Y < r\alpha/2\}, \tag{523.11}$$

with value zero when no such j exists. The balls in (523.11) are disjoint, and this is a Borel integer output. If $x = Z + rv_j$ and $\|Sx - A(x)\|_Y < \tau$, then (523.11) returns j. The same tightness and averaging argument as in (523.6)–(523.7) gives error at least τ. Let $\rho \downarrow 0$ and $r \uparrow 1$.

Restriction to E preserves all required measurability. In particular, $S(E)$ is separable, so the norm error is jointly measurable even when Y is nonseparable: approximate Sx by

countably valued Borel maps and take limits of the corresponding norm differences. Expected budgets are handled without any projection of the Y-valued output:

$$\mathbb{E}\left\|Sx - A^{(k)}(x)\right\|_Y^2 \le e^2 + \|S\|^2\, n/k. \tag{523.12}$$

Thus the lower bound $\alpha/2$ survives expected cardinality.

Conversely, suppose that $K = \overline{S(B_X)}$ is compact. Fix $\varepsilon > 0$. For each $z \in K - K$ with $\|z\|_Y \ge \varepsilon$, choose a norm-one functional $y'_z \in Y'$ with $y'_z(z) = \|z\|_Y$. Compactness gives finitely many of these functionals, $y'_1, \dots, y'_m$, such that

$$z, w \in K, \quad \max_i |y'_i(z - w)| \le \varepsilon/2 \quad \Longrightarrow \quad \|z - w\|_Y < \varepsilon. \tag{523.13}$$

Indeed, the open sets $\{q : |y'_z(q)| > \varepsilon/2\}$ cover the compact subset $\{q \in K - K : \|q\|_Y \ge \varepsilon\}$.

Measure the fixed functionals $L_i = y'_i \circ S$, and take a finite $\varepsilon/4$-net $z_1, \dots, z_N$ in K. Given the data $t = (t_i)$, return the first z_j satisfying

$$\max_i |t_i - y'_i(z_j)| \le \varepsilon/4, \tag{523.14}$$

and return zero outside the union of these finitely many boxes. For actual data, such a center exists; (523.13) gives error less than ε. This decoder is Borel and has finite range. If the compact set used in (523.13) is empty, a constant decoder already suffices. Hence every desired accuracy is achieved by deterministic nonadaptive continuous-linear information. No approximation property of X or Y is needed. □

Corollary 1.6 (A quantitative lower bound for separable codomains)**.** *Under the hypotheses of Corollary 1.5, suppose also that Y is separable. Define the Hausdorff noncompactness radius by*

$$\chi(S(B_X)) = \inf\{r > 0 : S(B_X) \text{ is covered by finitely many open } Y\text{-balls of radius } r\}.$$

Then every finite expected budget satisfies

$$e_n(S) \ge \chi(S(B_X)).$$

Proof. Assume $\chi(S(B_X)) > 0$, and fix $0 < r < \chi(S(B_X))$. Choose a countable $V \subset B_X$ such that $S(V)$ is dense in $S(B_X)$, and let E be its closed linear span in X. Use a full-support Gaussian Z on E, conditioned on $\|Z\|_X \le \rho$. Fix $0 < \theta < 1$ with $\theta + \rho < 1$, and use the bounded countable shifts θV.

Let (y_i) be dense in Y. For $\varepsilon > 0$, set

$$R_\varepsilon(y) = \min\{i \ge 1 : \|y - y_i\|_Y < \varepsilon\}.$$

This is a finite Borel integer. For a strict finite-query algorithm, Lemma 1.2 applied to $R_\varepsilon \circ A_\omega$ gives, for each seed,

$$q_M(\omega) := \sup_{v \in V} \mathbb{P}_Z(R_\varepsilon(A_\omega(\theta v + Z)) > M \mid \|Z\|_X \le \rho) \longrightarrow 0.$$

Countability of V and joint output measurability make q_M measurable in the seed. Thus $\mathbb{E}_\omega q_M \to 0$.

For each M, the definition of χ and the density of $S(V)$ give $v_M \in V$ with

$$\min_{i \le M} \|\theta S v_M - y_i\|_Y > \theta r.$$

Indeed, failure would give a finite closed-ball cover of $S(B_X)$ with radius r, and hence open-ball covers of every larger radius, contrary to $r < \chi$. On $R_\varepsilon(A_\omega(\theta v_M + Z)) \le M$, the error is at least $\theta r - \|S\|\,\rho - \varepsilon$. Its probability tends to one by the uniform averaged bound above. The worst-input RMS error is consequently at least this number. Let $\rho, \varepsilon \downarrow 0$, then $\theta \uparrow 1$, and finally $r \uparrow \chi(S(B_X))$.

For expected budgets, truncate after k queries and use (523.12). The strict-budget lower bound gives $e^2 + \|S\|^2\, n/k \ge \chi(S(B_X))^2$; let $k \to \infty$. □

References

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